Computing & Programming Foundations

⏱️ Duration: 2 Hours
🎯 Audience: Absolute Beginners
📌 Outcome: Comfort with computers + tool awareness + learning readiness

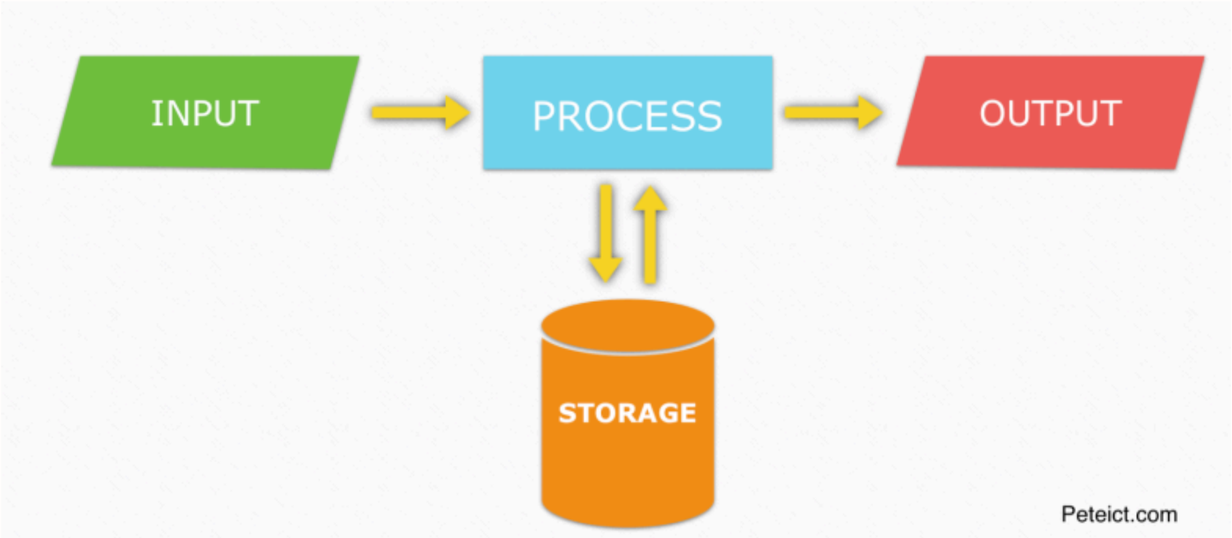
Recap

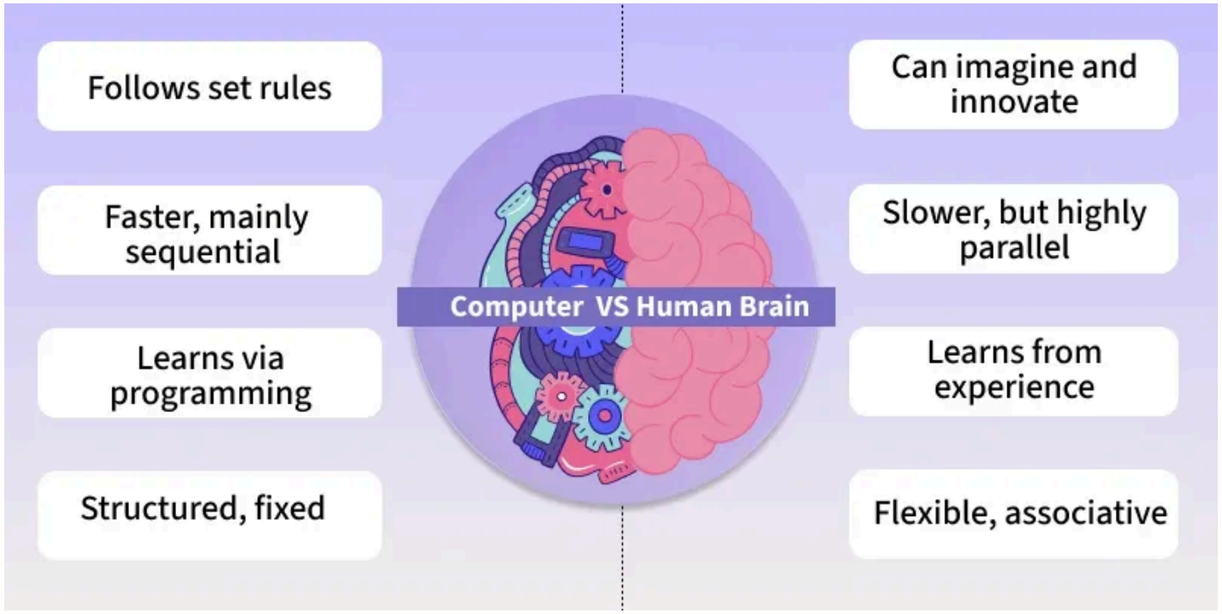
Welcome + Why This Course
Traditional Software vs AI
AI vs ML vs Deep Learning vs GenAI
History of AI (Failures → Breakthroughs)
Current Industry Use Cases
Career Landscape & Roles
Course Roadmap + What’s Coming Next

Agenda

Duration	Session Title
15 min	What Is a Computer?
15 min	Hardware vs Software
15 min	What Is a Kernel?
15 min	Software, Program & Code
15 min	Data & Logic
15 min	Traditional Software
15 min	IDE & Terminal
15 min	Files & File Type
15 min	How code runs

WHAT IS A COMPUTER?

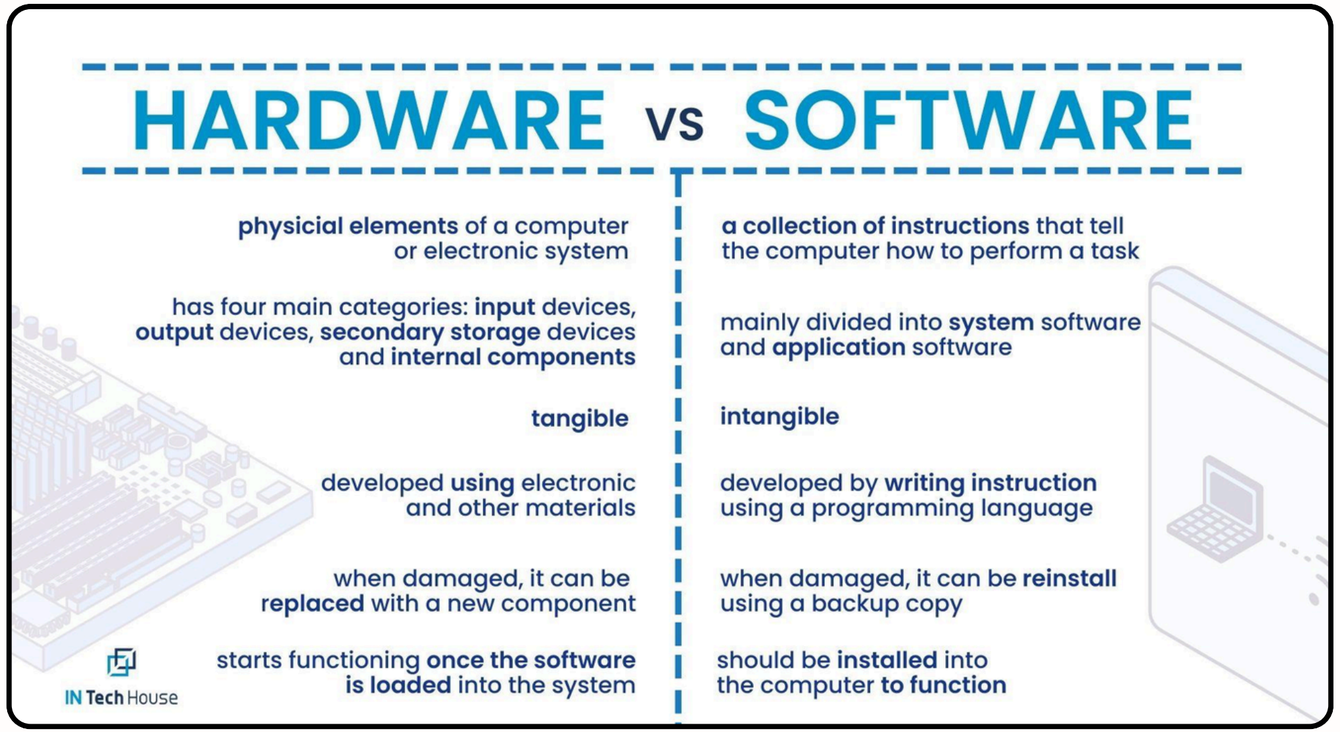




A computer only does 3 things:

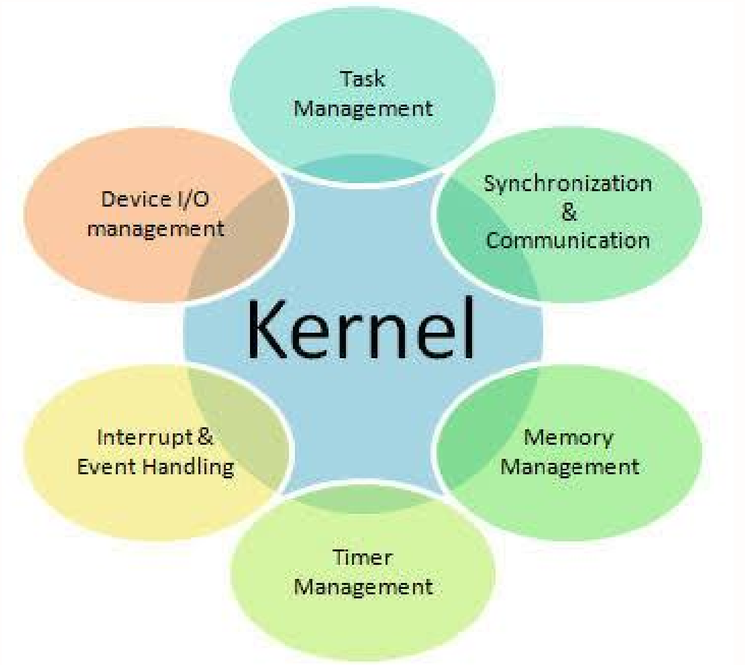
- Input – you give something
- Process – it follows instructions
- Output – it shows result

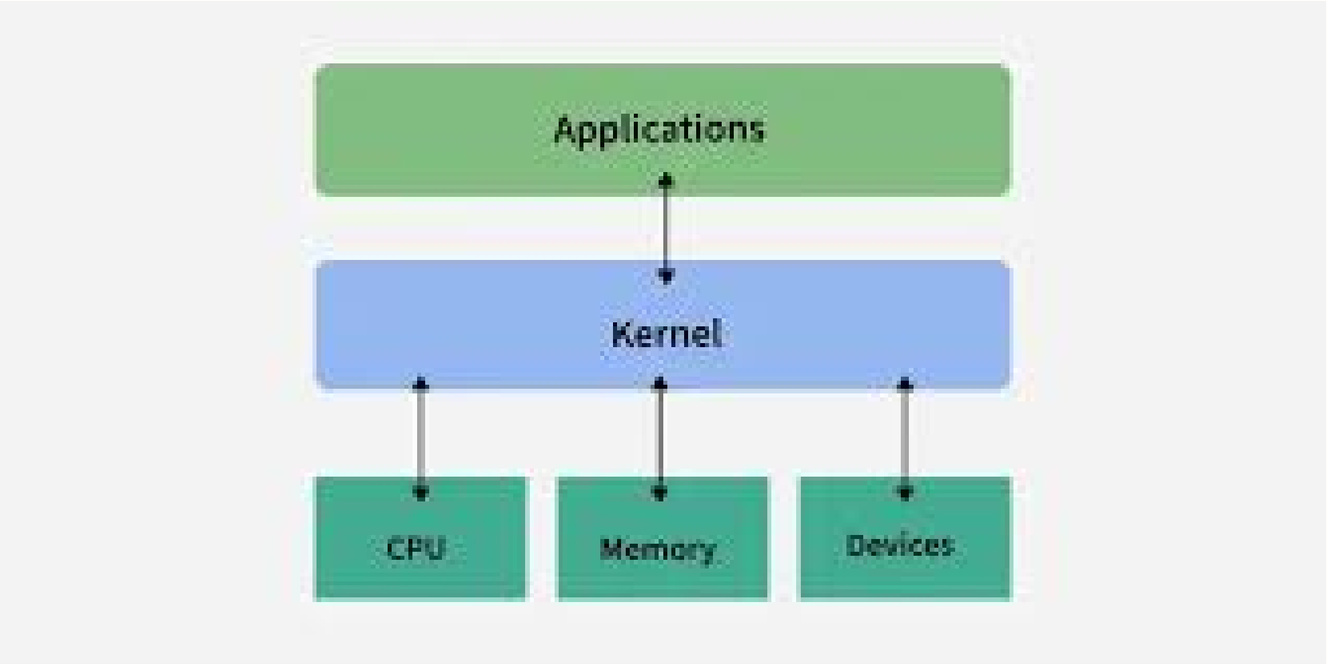
HARDWARE vs SOFTWARE



Hardware	**Software**
Keyboard	WhatsApp
Mouse	Chrome
Screen	Calculator
CPU	Music player

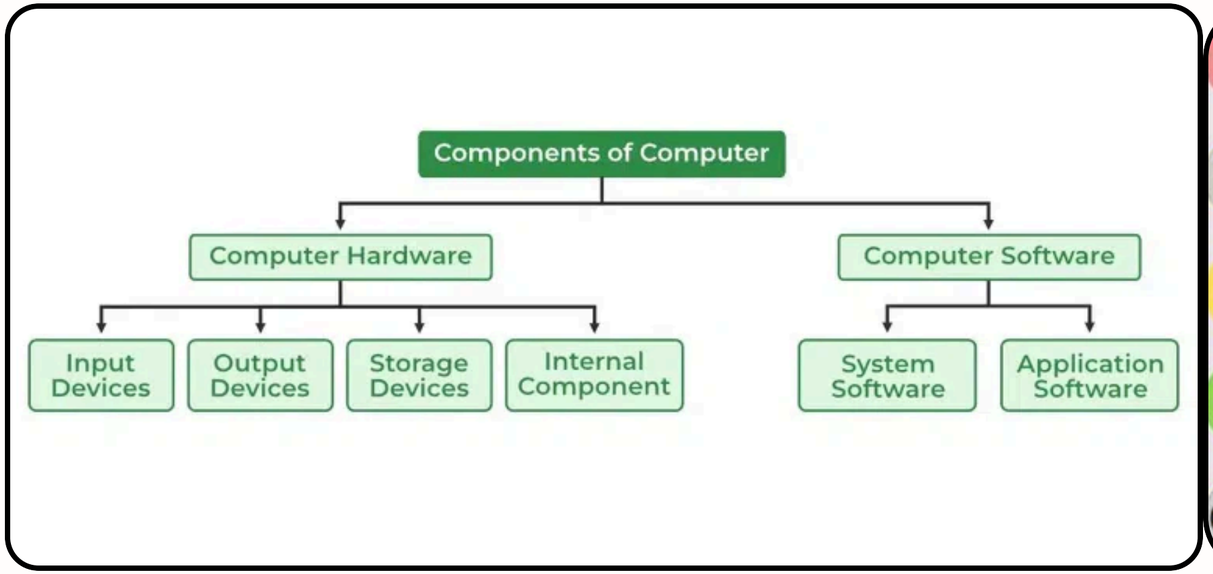
WHAT IS KERNEL?

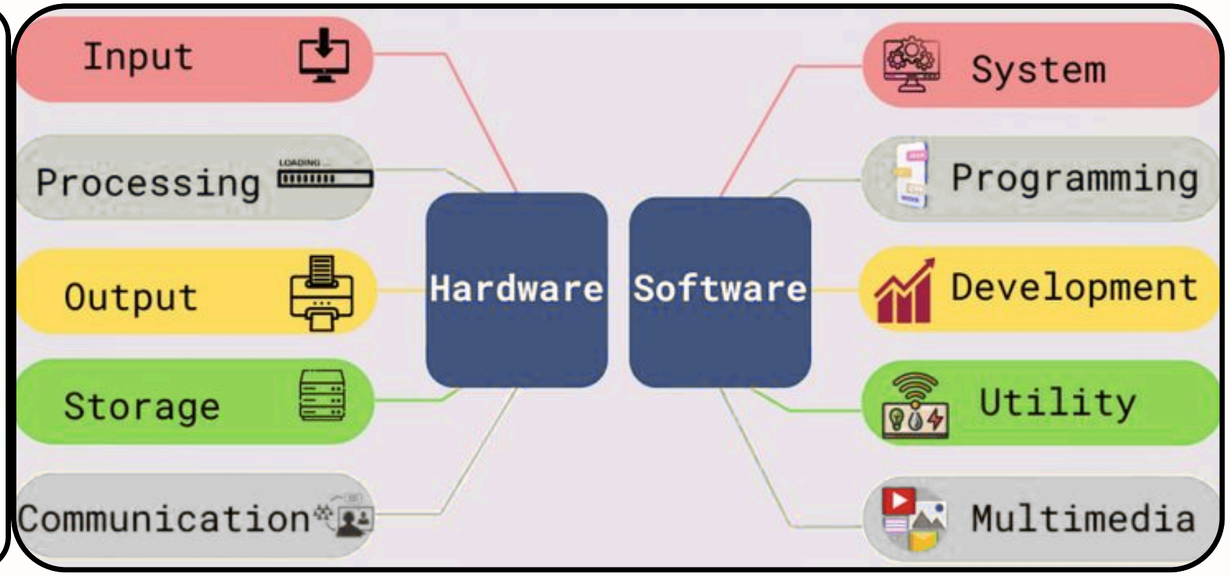




The kernel handles essential tasks to keep your computer humming along smoothly.
It manages memory, ensuring that programs get the resources they need.

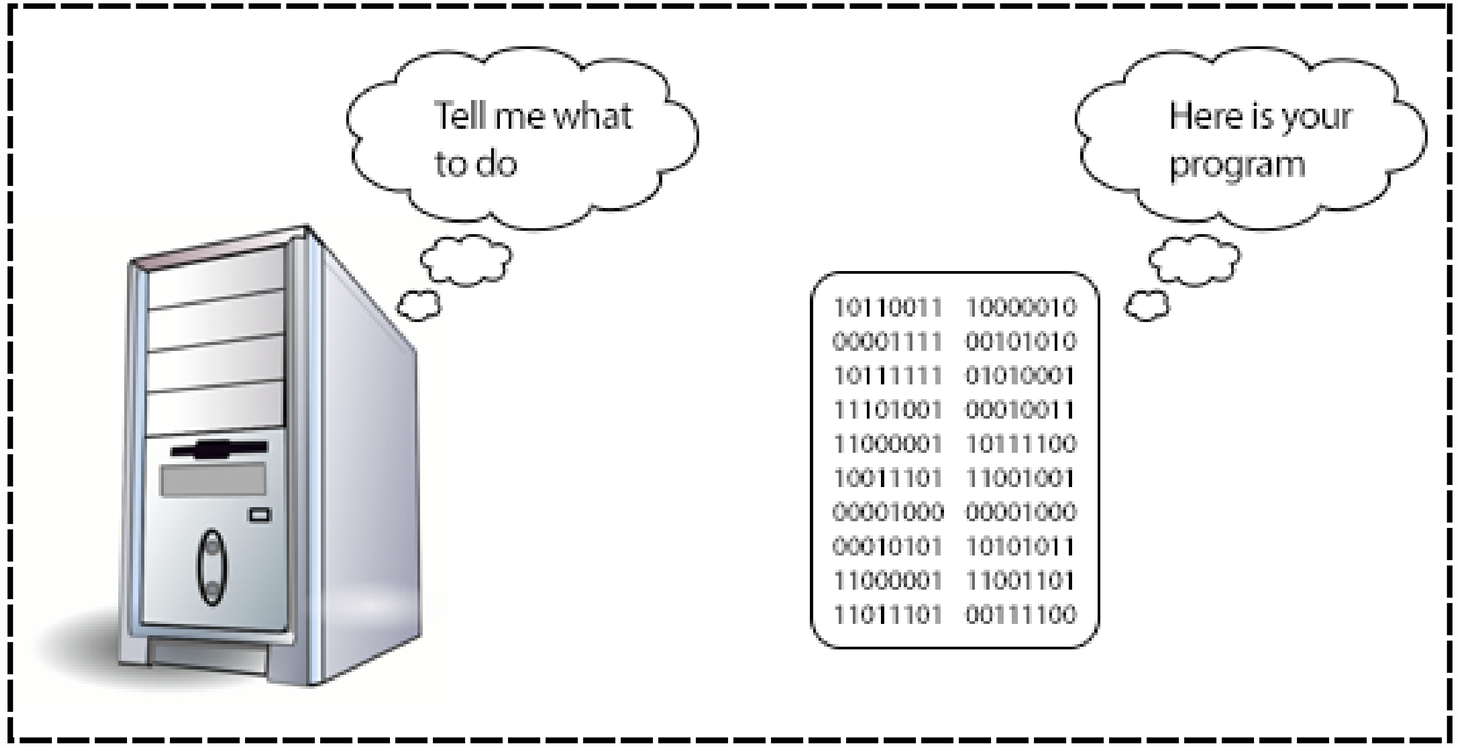
WHAT IS SOFTWARE?





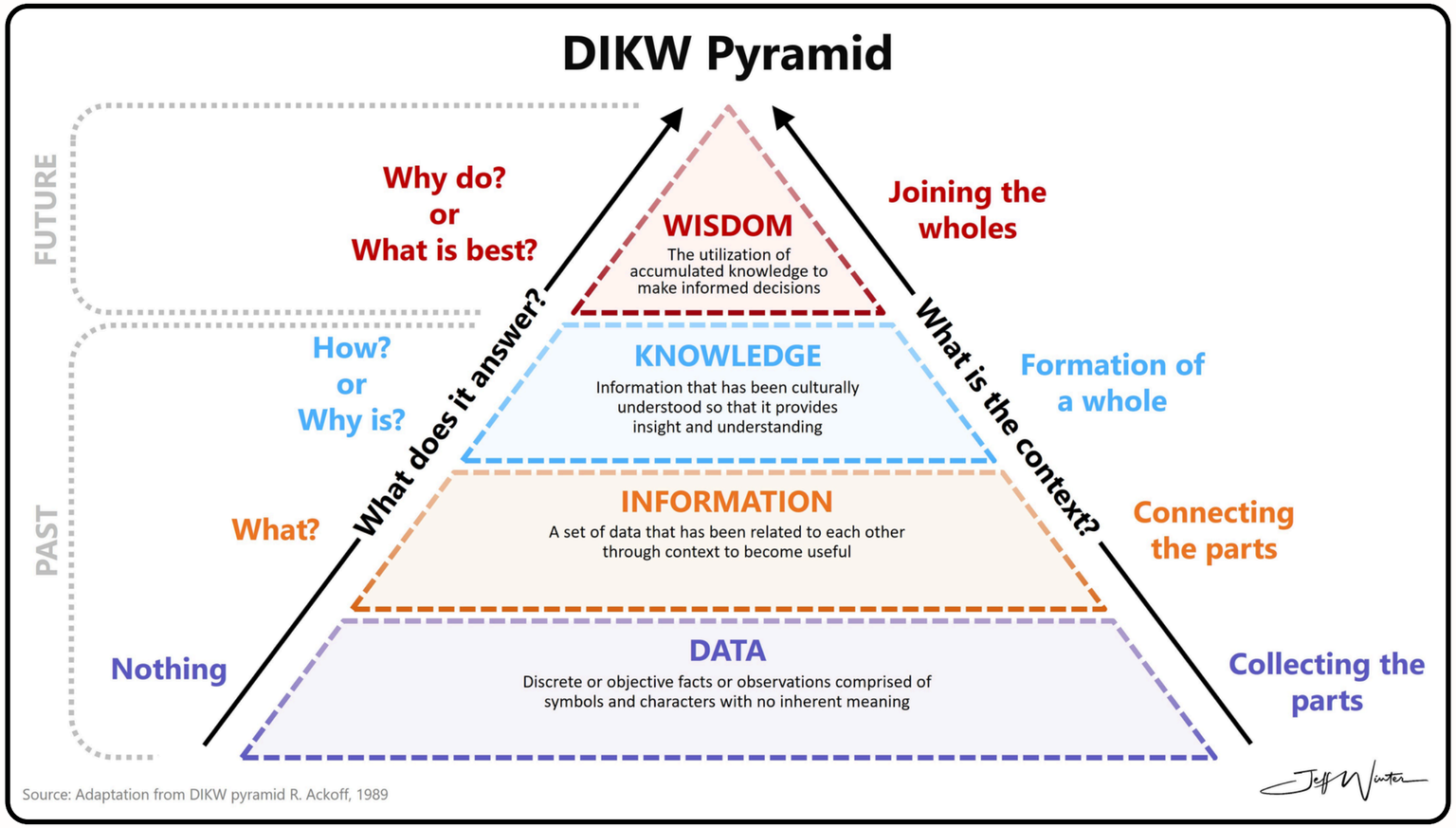
Think of a computer as the body and the software as its brain.....🧠

WHAT IS PROGRAM?



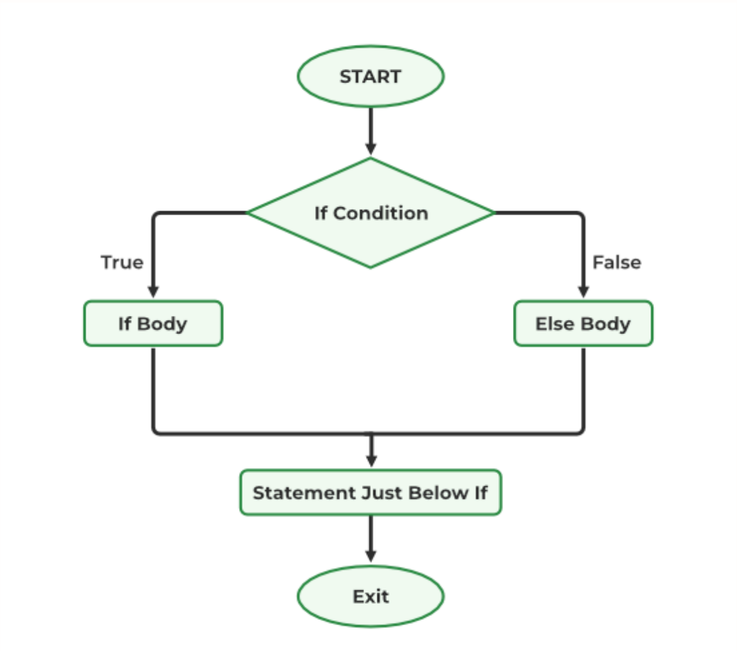
A program is a written instruction note for a computer....📝.

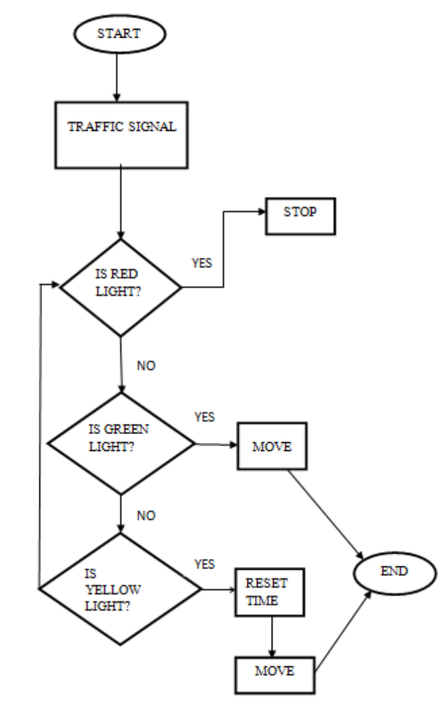
WHAT IS DATA?



Examples: Marks, Height, Age, Messages, Photos.

WHAT IS LOGIC?



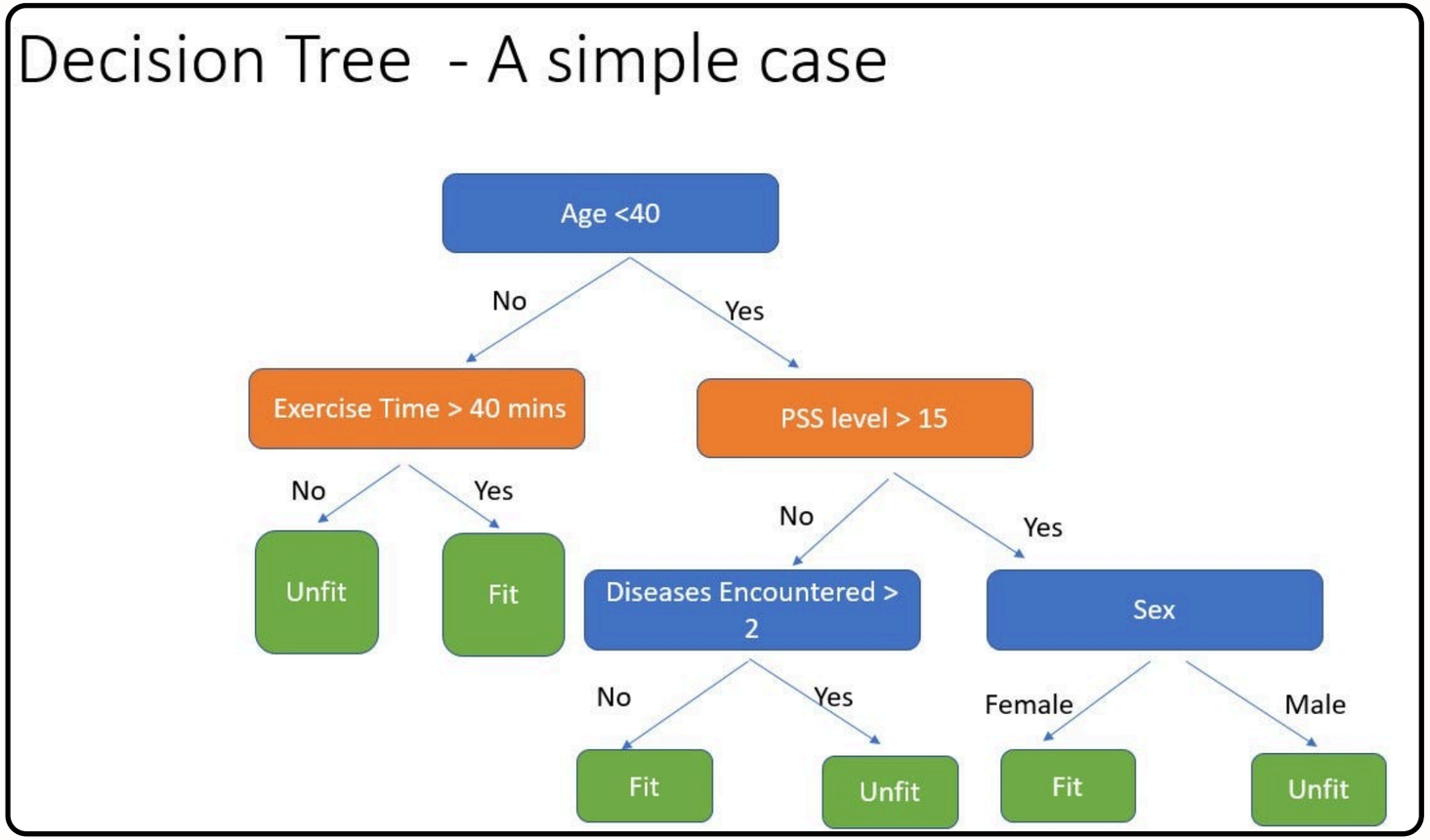


Logic means making a decision based on a condition.

In very simple words:

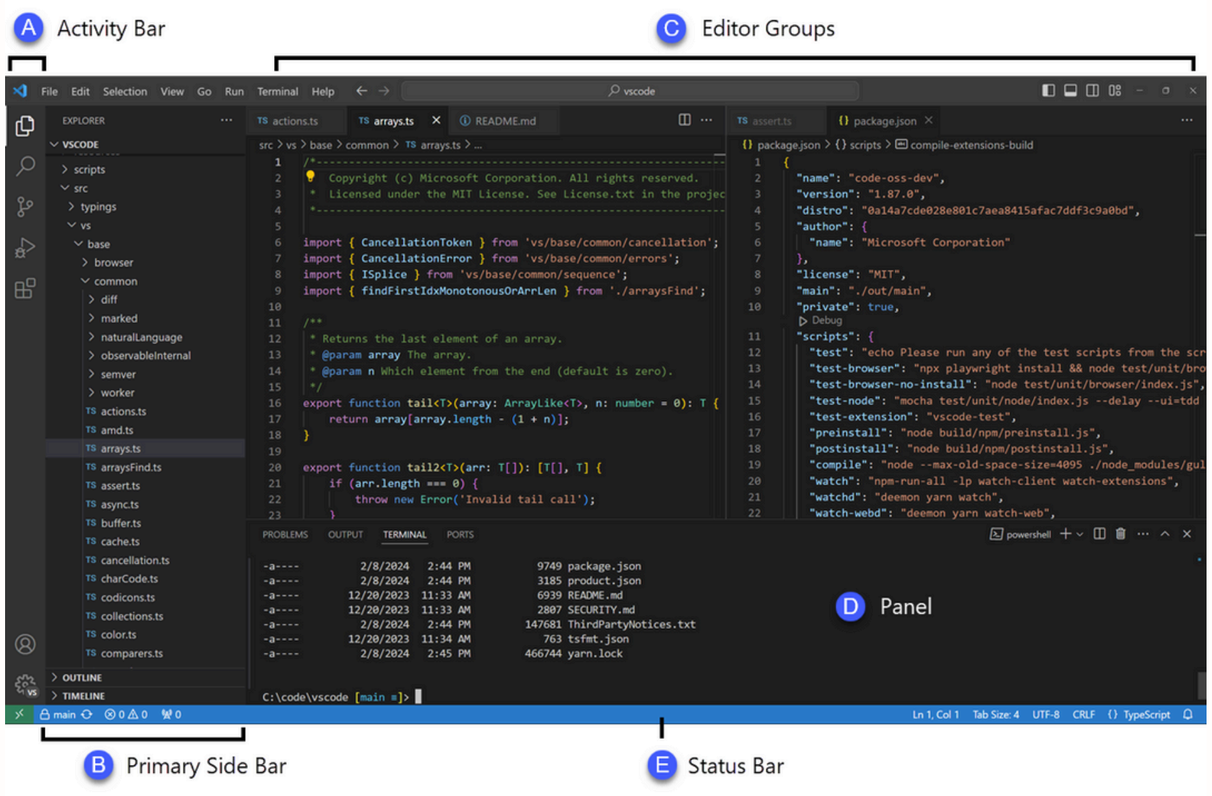
- Something happens
- You decide what to do
- Based on a rule

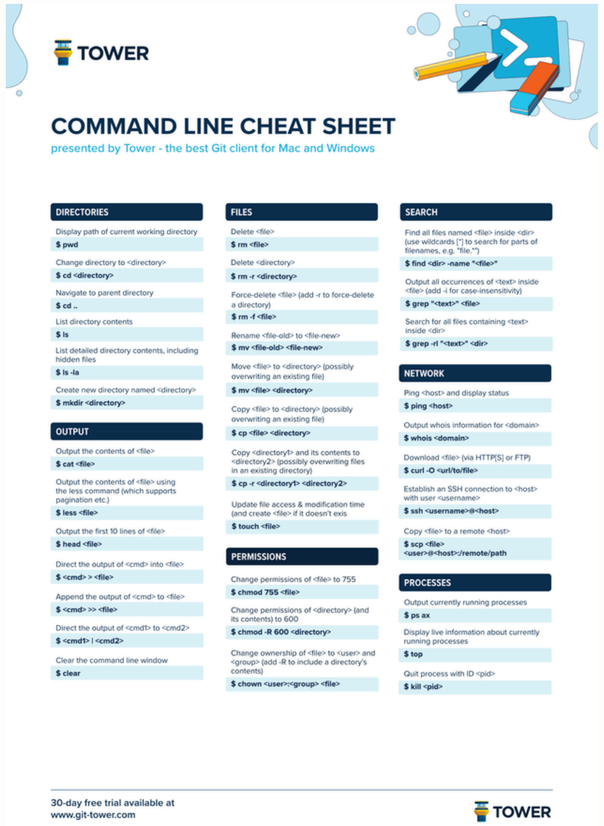
TRADITIONAL SOFTWARE



Traditional software works only by fixed rules written by humans.
It does not learn, does not change, and does not improve.

WHAT IS IDE & TERMINAL (CLI)?





IDE is a workplace for writing code.

Examples:

- VS Code
- PyCharm

Terminal is a text-based way to talk to computer.

WHAT IS FILES, FILE TYPES?

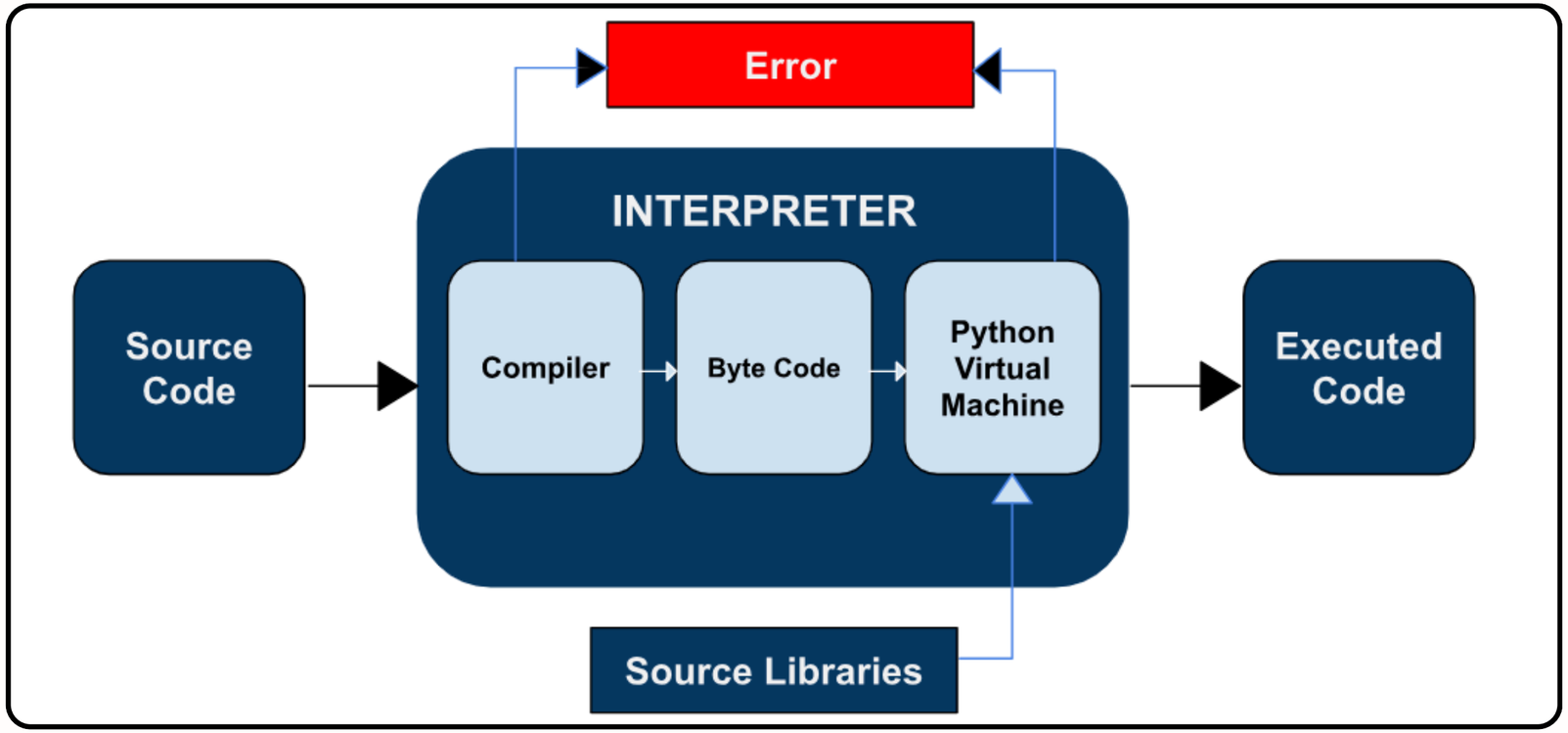
File Types (VERY IMPORTANT)

Type	Meaning
`.txt`	Text
`.jpg`	Image
`.mp3`	Audio
`.mp4`	Video
`.py`	Python code

File = Document
Examples:
Photo file
Song file
Video file

Folder = Bag of files

HOW CODE RUNS?



- You write code in a .py file
- Python reads it line by line
- Computer executes instructions
- Output appears on screen

THANK YOU